Masaki Ogura

Curriculum Vitae

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AREAS OF EXPERTISE

Control theory, network science, optimization, stochastic processes, biological physics

EDUCATION

- Ph.D. (Mathematics), *Texas Tech University*. 2014
- M.Sc. (Informatics), *Kyoto University*. 2009
- B.Eng., *Kyoto University*. 2007

PROFESSIONAL APPOINTMENTS

- Professor, Graduate School of Advanced Science and Engineering, Hiroshima University, Japan. Apr 2024–present
- Visiting Professor, Department of Bioinformatic Engineering, Graduate School of Information Science and Technology, Osaka University, Japan. Apr 2024–present
- Associate Professor, Department of Bioinformatic Engineering, Graduate School of Information Science and Technology, Osaka University, Japan. Nov 2019–Mar 2024
- Visiting Associate Professor, Division of Information Science, Graduate School of Science and Technology, Nara Institute of Science and Technology, Japan. Apr 2020–Mar 2021
- Assistant Professor, Graduate School of Information Science, Nara Institute of Science and Technology, Japan. Mar 2017–Oct 2019
- Postdoctoral Researcher, Department of Electrical and Systems Engineering, University of Pennsylvania, Oct 2014–Feb 2017

SHORT TERM VISITS

- Department of Mechanical Engineering, University of Hong Kong. 2018, 2019
- ICTEAM Institute, Université catholique de Louvain, Belgium. 2013

SELECTED AWARDS AND HONORS

- Runner-up of the 2019 Best Paper Award, *IEEE Transactions on Network Science and Engineering*. 2019
- Summer Dissertation/Thesis Research Award, Texas Tech University. 2014
- Cash Family Endowed Fellowship, Texas Tech University. 2013

- Best Paper Award, The Society of Instrument and Control Engineers. 2012

PUBLICATIONS

Book Chapters

[1] K. Sakurama, K. Kashima, T. Ikeda, N. Hayashi, K. Hoshino, M. Ogura, and C. Zhao, "System-Control-Based Approach to Car-Sharing Systems," in *Advanced Mathematical Science for Mobility Society*. Springer Singapore, 2024, pp. 127–171.

[2] V. M. Preciado, M. Zargham, C. Nowzari, S. Han, M. Ogura, A. Jadbabaie, and G. J. Pappas, "Bio-inspired Framework for Allocation of Protection Resources in Cyber-Physical Networks," in *Principles of Cyber-Physical Systems*. Cambridge University Press, 2020, pp. 293–322.

[3] M. Ogura and V. M. Preciado, "Optimal Containment of Epidemics in Temporal and Adaptive Networks," in *Temporal Networks Epidemiology*. Springer, 2017, pp. 241–266.

[4] M. Ogura and C. F. Martin, "Linear Switching Systems and Random Products of Matrices," in *Mathematical System Theory – Festschrift in Honor of Uwe Helmke on the Occasion of his Sixtieth Birthday*, K. Hüper and J. Trumpf, Eds. CreateSpace, 2013, pp. 291–300.

Refereed Journal Articles

[1] Y. Chen, H. Zhao, M. Ogura, Y. Gao, and L. Peng, "Data-driven dual-channel dynamic event-triggered load frequency control for multiarea power systems under DoS attacks," *Energy Science & Engineering*, vol. 14, no. 2, pp. 780–792, 2026.

[2] Q. Shi, M. Ogura, Q. Wang, F. Cheng, and L. Peng, "Leading cruise control under human reaction delays: stabilizability, string stability, and mixed H_2/H_∞ synthesis," *IEEE Transactions on Automation Science and Engineering* (accepted for publication), 2026.

[3] Y. Higuchi, R. Nagai, Y. Bai, A. Okazaki, N. Wakamiya, and M. Ogura, "Development and evaluation of a black-box optimization framework for weather-intervention design," *Journal of Computational Science*, vol. 97, p. 102850, 2026.

[4] H. Maeda, Y. Bai, N. Wakamiya, and M. Ogura, "Modeling and analysis of locust hoppers dynamics under olfactory receptor 35 deficiency," *Ecological Modelling*, vol. 517, p. 111608, 2026.

[5] S. Yang, L. Peng, M. Ogura, and Y. Chen, "Dual-channel dynamic event-triggered data-driven control for energy storage system using quantised information," *Journal of Control and Decision* (accepted for publication), 2026.

[6] Q. Wang, Y. Bai, M. Ogura, and L. Peng, "Periodic event-triggered generalized dissipative state estimation for hidden semi-Markov jump systems," *ISA Transactions*, vol. 173, pp. 122–130, 2026.

[7] S. Yang, L. Peng, M. Ogura, Y. Gao, and Y. Chen, "Dual-channel event-triggered data-driven control for energy storage system with input saturation and packet loss," *Journal of Automation and Intelligence* (accepted for publication), 2026.

[8] Y. Chen, H. Zhao, M. Ogura, H. Yu, and L. Peng, "Data-driven event-triggered fixed-time load frequency control for multi-area power systems with input delays," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 72, no. 12, pp. 8492–8504, 2025.

[9] Y. Chen, H. Zhao, M. Ogura, Y. Gao, and L. Peng, "Data-driven dual-channel dynamic event-triggered load frequency control for multi-area power systems with uniform quantizer," *Science Progress*, vol. 108, no. 4, p. 00368504251391464, 2025.

[10] S. Yang, H. Zhao, M. Ogura, and L. Peng, "Data-driven adaptive optimal output tracking with assured convergence rate," *Journal of the Franklin Institute*, vol. 362, no. 16, p. 108086, 2025.

[11] Y. Deng, M. Ogura, Y. Bai, Y. Wang, and M. Feroskhan, "Safety-critical control for nonlinear systems with complex input constraints," *IEEE Transactions on Automatic Control*, vol. 70, no. 10, pp. 7016–7023, 2025.

[12] F. Sato, M. Ogura, A. Sashie, Y. Bai, M. Shimono, and N. Wakamiya, "Neural SDE-based spike control of noisy neurons," *PLOS One*, vol. 20, no. 9, p. e0330607, 2025.

[13] R. Nagai, Y. Bai, M. Ogura, S. Kotsuki, and N. Wakamiya, "Evaluation of effectiveness of intervention strategy in control simulation experiment through comparison with model predictive control," *Nonlinear Processes in Geophysics*, vol. 32, no. 3, pp. 281–292, 2025.

[14] T. Ohtsuka, A. Okazaki, M. Ogura, and S. Kotsuki, "Convex optimization of initial perturbations toward quantitative weather control," *Scientific Online Letters on the Atmosphere*, vol. 21, pp. 2025–020, 2025.

[15] A. Fujioka, M. Ogura, and N. Wakamiya, "Cyclic pursuit formation control for arbitrary desired shapes," *Journal of the Franklin Institute*, vol. 362, no. 2, p. 107467, 2025.

[16] Y. Bai, P. T. T. Ngoc, H. D. Nguyen, D. L. Le, Q. H. Ha, K. Kai, Y. X. S. To, Y. Deng, J. Song, N. Wakamiya, H. Sato, and M. Ogura, "Swarm navigation of cyborg-insects in unknown obstructed soft terrain," *Nature Communications*, vol. 16, p. Article number: 221, 2025.

[17] A. Li, M. Ogura, and N. Wakamiya, "Swarm shepherding using bearing-only measurements," *Philosophical Transactions of the Royal Society A*, vol. 383, no. 2289, p. 20240145, 2025.

[18] J. Aizawa, M. Ogura, M. Shimono, and N. Wakamiya, "Firing pattern manipulation of neuronal networks by deep unfolding-based model predictive control," *IET Control Theory Applications*, vol. 18, no. 15, pp. 2003–2013, 2024.

[19] C. Zhao, B. Zhu, M. Ogura, and J. Lam, "Probability rate optimization of positive Markov jump linear systems via DC programming," *Asian Journal of Control*, vol. 26, no. 5, pp. 2242–2249, 2024.

[20] W. Imahayashi, Y. Tsunoda, and M. Ogura, "Route design in sheepdog system–traveling salesman problem formulation and evolutionary computation solution–," *Advanced Robotics*, vol. 38, no. 9-10, pp. 632–646, 2024.

[21] J. J. R. Liu, M. Ogura, Q. Li, and J. Lam, "Learning-based stabilization of Markov jump linear systems," *Neurocomputing*, vol. 584, p. 127618, 2024.

[22] R. Liu, M. Ogura, E. F. Dos Reis, and N. Masuda, "Effects of concurrency on epidemic spreading in Markovian temporal networks," *European Journal of Applied Mathematics*, vol. 35, no. 3, pp. 430–461, 2024.

[23] M. Ogura and C. F. Martin, "Mean escape time of switched Riccati differential equations," *Journal of the Franklin Institute*, vol. 360, no. 10, pp. 6827–6845, 2023.

[24] C. Zhao, K. Sakurama, and M. Ogura, "Optimization of buffer networks via DC programming," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 70, no. 2, pp. 606–610, 2023.

[25] M. Ogura and N. Wakamiya, "Reduced-order model predictive control of a fish schooling model," *Nonlinear Analysis: Hybrid Systems*, vol. 49, p. 101342, 2023.

[26] C. Zhao, B. Zhu, M. Ogura, and J. Lam, "Parameterized synthesis of discrete-time positive linear systems: A geometric programming perspective," *IEEE Control Systems Letters*, vol. 7, pp. 2551–2556, 2023.

[27] X. Gong, M. Ogura, J. Shen, T. Huang, and Y. Cui, "Optimal epidemics policy seeking on networks-of-networks under malicious attacks by geometric programming," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 6, pp. 3845–3857, 2023.

[28] M. Ogura and C. Nowzari, "Special issue: modeling, analysis, and control of epidemics in networks," *Mathematical Biosciences and Engineering*, vol. 20, no. 5, pp. 8446–8447, 2023.

[29] A. Li, M. Ogura, and N. Wakamiya, "Communication-free shepherding navigation with multiple steering agents," *Frontiers in Control Engineering*, vol. 4, 2023.

[30] A. Fujioka, M. Ogura, and N. Wakamiya, "Shepherding algorithm for heterogeneous flock with model-based discrimination," *Advanced Robotics*, vol. 37, no. 1-2, pp. 99–114, 2023.

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[32] M. Nagahara, M. Ogura, and Y. Yamamoto, "Iterative greedy LMI for sparse control," *IEEE Control Systems Letters*, vol. 6, pp. 986–991, 2022.

[33] M. Kumazaki, M. Ogura, and T. Tachibana, "Dynamic service chain construction based on model predictive control in NFV environments," *IEICE Transactions on Communications*, vol. E105-B, no. 4, pp. 399–410, 2022.

[34] M. Kishida* and M. Ogura*, "Temporal deep unfolding for constrained nonlinear stochastic optimal controls," *IET Control Theory & Applications*, vol. 16, no. 2, pp. 139–150, 2022. (*equal contribution)

[35] R. Himo, M. Ogura, and N. Wakamiya, "Iterative algorithm for shepherding unresponsive sheep," *Mathematical Biosciences and Engineering*, vol. 19, no. 4, pp. 3509–3525, 2022.

[36] T. Kimura and M. Ogura, "Distributed 3D deployment of aerial base stations for on-demand communication," *IEEE Transactions on Wireless Communications*, vol. 20, no. 12, pp. 7728–7742, 2021.

[37] K. Hashimoto, Y. Onoue, M. Ogura, and T. Ushio, "Event-triggered control for mitigating SIS spreading processes," *Annual Reviews in Control*, vol. 52, pp. 479–494, 2021.

[38] K. Sugimoto, T. Aihara, M. Ogura, and K. Hanada, "Gain scheduling for sampled-data state estimation over lossy networks," *Transactions of the Institute of Systems, Control and Information Engineers*, vol. 34, no. 11, pp. 287–293, 2021.

[39] M. Nagahara, B. Krishnamachari, M. Ogura, A. Ortega, Y. Tanaka, Y. Ushifusa, and T. W. Valente, "Control, intervention, and behavioral economics over human social networks against COVID-19," *Advanced Robotics*, vol. 35, no. 11, pp. 733–739, 2021.

[40] C. Zhao, M. Ogura, M. Kishida, and A. Yassine, "Optimal resource allocation for dynamic product development process via convex optimization," *Research in Engineering Design*, vol. 32, no. 1, pp. 71–90, 2021.

[41] X. Chen, M. Ogura, and V. M. Preciado, "SDP-Based moment closure for epidemic processes on networks," *IEEE Transactions on Network Science and Engineering*, vol. 7, no. 4, pp. 2850–2865, 2020.

[42] M. Ogura, M. Kishida, and J. Lam, "Geometric programming for optimal positive linear systems," *IEEE Transactions on Automatic Control*, vol. 65, no. 11, pp. 4648–4663, 2020.

[43] W. Mei, C. Zhao, M. Ogura, and K. Sugimoto, "Mixed H_2/H_∞ control of delayed Markov jump linear systems," *IET Control Theory Applications*, vol. 14, no. 15, pp. 2076–2083, 2020.

[44] C. Zhao, M. Ogura, and K. Sugimoto, "Stability optimization of positive semi-Markov jump linear systems via convex optimization," *SICE Journal of Control, Measurement, and System Integration*, vol. 13, no. 5, pp. 233–239, 2020.

[45] M. Kishida*, M. Ogura*, Y. Yoshida, and T. Wadayama, "Deep learning-based average consensus," *IEEE Access*, vol. 8, pp. 142 404–142 412, 2020. (*equal contribution)

[46] N. Masuda, V. M. Preciado, and M. Ogura, "Analysis of the susceptible-infected-susceptible epidemic dynamics in networks via the non-backtracking matrix," *IMA Journal of Applied Mathematics*, vol. 85, no. 2, pp. 214–230, 2020.

[47] X. Chen, M. Ogura, and V. M. Preciado, "Bounds on the spectral radius of digraphs from motif counts," *SIAM Journal on Matrix Analysis and Applications*, vol. 41, no. 2, pp. 525–553, 2020.

[48] M. Ogura and V. M. Preciado, "Stability of SIS spreading processes in networks with non-Markovian transmission and recovery," *IEEE Transactions on Control of Network Systems*, vol. 7, no. 1, pp. 349–359, 2020.

[49] M. Ogura, W. Mei, and K. Sugimoto, "Synergistic effects in networked epidemic spreading dynamics," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 67, no. 3, pp. 496–500, 2020.

[50] Y. Abe, M. Ogura, H. Tsuji, A. Miura, and S. Adachi, "Resource and network management framework for a large-scale satellite communications system," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E103, no. 2, pp. 492–501, 2020. **Telecom System Technology Student Award**, Telecommunication Advancement Foundation.

[51] M. Ogura, J. Harada, M. Kishida, and A. Yassine, "Resource optimization of product development projects with time-varying dependency structure," *Research in Engineering Design*, vol. 30, no. 3, pp. 435–452, 2019.

[52] M. Ogura, V. M. Preciado, and N. Masuda, "Optimal containment of epidemics over temporal activity-driven networks," *SIAM Journal on Applied Mathematics*, vol. 79, no. 3, pp. 986–1006, 2019.

[53] W. Mei and M. Ogura, "Kronecker weights for instability analysis of Markov jump linear systems," *IET Control Theory & Applications*, vol. 13, no. 3, pp. 360–366, 2019.

[54] M. Wakaiki, M. Ogura, and J. P. Hespanha, "LQ-optimal sampled-data control under stochastic delays: gridding approach for stabilizability and detectability," *SIAM Journal on Control and Optimization*, vol. 56, no. 4, pp. 2634–2661, 2018.

[55] M. Ogura and V. M. Preciado, "Second-order moment-closure for tighter epidemic thresholds," *Systems & Control Letters*, vol. 113, pp. 59–64, 2018.

[56] M. Ogura, A. Cetinkaya, T. Hayakawa, and V. M. Preciado, "State feedback control of Markov jump linear systems with hidden-Markov mode observation," *Automatica*, vol. 89, pp. 65–72, 2018.

[57] M. Ogura and V. M. Preciado, "Optimal design of switched networks of positive linear systems via geometric programming," *IEEE Transactions on Control of Network Systems*, vol. 4, no. 2, pp. 213–222, 2017.

[58] M. Ogura, M. Wakaiki, H. Rubin, and V. M. Preciado, "Delayed bet-hedging resilience strategies under environmental fluctuations," *Physical Review E*, vol. 95, p. 052404, 2017.

[59] M. Ogura, V. M. Preciado, and R. M. Jungers, "Efficient method for computing lower bounds on the p-radius of switched linear systems," *Systems & Control Letters*, vol. 94, pp. 159–164, 2016.

[60] M. Ogura and V. M. Preciado, "Epidemic processes over adaptive state-dependent networks," *Physical Review E*, vol. 93, p. 062316, 2016.

[61] M. Ogura and V. M. Preciado, "Stability of Markov regenerative switched linear systems," *Automatica*, vol. 69, pp. 169–175, 2016.

[62] M. Ogura and V. M. Preciado, "Stability of spreading processes over time-varying large-scale networks," *IEEE Transactions on Network Science and Engineering*, vol. 3, no. 1, pp. 44–57, 2016. Runner-up of **2019 IEEE TNSE Best Paper Award**.

[63] M. Ogura and C. F. Martin, "Stability analysis of linear systems subject to regenerative switchings," *Systems & Control Letters*, vol. 75, pp. 94–100, 2015.

[64] M. Ogura and C. F. Martin, "A limit formula for joint spectral radius with p-radius of probability distributions," *Linear Algebra and its Applications*, vol. 458, pp. 605–625, 2014.

[65] M. Ogura and C. F. Martin, "Stability analysis of positive semi-Markovian jump linear systems with state resets," *SIAM Journal on Control and Optimization*, vol. 52, pp. 1809–1831, 2014.

[66] M. Ogura and C. F. Martin, "Generalized joint spectral radius and stability of switching systems," *Linear Algebra and its Applications*, vol. 439, no. 8, pp. 2222–2239, 2013.

[67] M. Ogura and Y. Yamamoto, "Dissipativity of pseudorational behaviors," *IEEE Transactions on Automatic Control*, vol. 58, no. 4, pp. 823–833, 2013.

[68] M. Nagahara, M. Ogura, and Y. Yamamoto, "H^∞ design of periodically nonuniform interpolation and decimation for non-band-limited signals," *SICE Journal of Control, Measurement, and System Integration*, vol. 4, no. 5, pp. 341–348, 2011. **2012 SICE Best Paper Award**.

Refereed Conference Proceedings

[1] C. Zhao, K. Sakurama, and M. Ogura, "Optimization of Markovian switching one-way car-sharing networks via DC programming," in *23rd IFAC World Congress* (accepted), 2026.

[2] T. Sone, M. Ogura, and M. Kishida, "Risk-aware model predictive control via temporal deep unrolling," in *2026 European Control Conference*, 2026, pp. 2858–2863.

[3] T. Sone, M. Ogura, and M. Kishida, "Temporal deep unrolling-based MPC for vehicle trajectory tracking," in *2025 IEEE International Conference on Systems, Man, and Cybernetics*, 2025, pp. 4534–4539.

[4] H. Maeda and M. Ogura, "Modeling and analysis of locust swarms with OR35-deficient individuals," in *9th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2025*, 2025.

[5] H. Maeda and M. Ogura, "Modeling and analysis of locust swarms with OR35-deficient individuals," in *2025 SICE Festival with Annual Conference*, 2025, pp. 168–170.

[6] A. Sashie and M. Ogura, "Effects of differences in swimming speed on heterogeneous fish schools in a set-net environment," in *2025 SICE Festival with Annual Conference*, 2025, pp. 171–174.

[7] T. Umezaki, M. Ogura, T. Nakamura, S. Suzuki, T. Takagi, T. Yoshikawa, S. Torisawa, K. Kato, Y. Washio, S. Masuma, and N. Wakamiya, "Modeling fish schooling behavior for selective fishing in set net fisheries," in *9th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2025*, 2025.

[8] Y. Higuchi, N. Rikuto, A. Okazaki, M. Ogura, and N. Wakamiya, "Comparative analysis of black-box optimization methods for weather intervention design," in *25th International Conference on Computational Science*, 2025, pp. 194–209.

[9] W. Imahayashi, T. Tokugawa, K. Koitabashi, Y. Tsunoda, M. Ogura, and S. Muto, "Actual experiment using mobile robot in sheepdog system -guidance route design using traveling salesman problem-," in *SICE Festival 2024 with Annual Conference*, 2024, pp. 165–168.

[10] J. Aizawa, M. Ogura, M. Shimono, and N. Wakamiya, "Manipulation of neuronal network firing patterns using temporal deep unfolding-based MPC," in *2023 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference*, 2023, pp. 15–21.

[11] Y. Deng, A. Li, M. Ogura, and N. Wakamiya, "Collision-free shepherding control of a single target within a swarm," in *2023 IEEE International Conference on Systems, Man, and Cybernetics*, 2023, pp. 5218–5223.

[12] A. Fujioka, M. Ogura, and N. Wakamiya, "Proposal and evaluation of pursuit formations based on cyclic pursuit dynamics," in *2023 International Symposium on Nonlinear Theory and its Applications*, 2023, pp. 30–33.

[13] M. Ogura and V. M. Preciado, "Asynchronous temporal interactions promote disparity in networks," in *2023 International Symposium on Nonlinear Theory and its Applications*, 2023, p. 25.

[14] A. Fujioka, M. Ogura, and N. Wakamiya, "Shepherding heterogeneous flocks: overview and prospect," in *22nd IFAC World Congress*, 2023, pp. 11 523–11 526.

[15] C. Zhao, X. Gong, Y. Ebihara, and M. Ogura, "Impulse-to-peak optimization of positive linear systems via DC programming," in *22nd IFAC World Congress*, 2023, pp. 5544–5549.

[16] Y. Deng, A. Li, M. Ogura, and N. Wakamiya, "Collision-free property analysis for the shepherding swarm," in *SICE Annual Conference 2023*, 2023, pp. 1072–1073.

[17] J. Aizawa, M. Ogura, M. Shimono, and N. Wakamiya, "Temporal deep unfolding-based MPC for controlling firing patterns of neuronal network," in *2023 RISP International Workshop on Nonlinear Circuits, Communications and Signal Processing*, 2023.

[18] A. Li, M. Ogura, and N. Wakamiya, "Proposal of a bearing-only shepherding algorithm with limited sensing capabilities," in *AROB-ISBC-SWARM 2023*, 2023, pp. GS34–1.

[19] Y. Deng, M. Ogura, A. Li, and N. Wakamiya, "Shepherding control for separating a single agent from a swarm," in *1st IFAC Workshop on Control of Complex Systems*, 2022, pp. 217–222.

[20] A. Fujioka, M. Ogura, and N. Wakamiya, "Shepherding algorithm based on variant agent detection for heterogeneous flock," in *SICE Annual Conference 2022*, 2022, pp. 87–91. Finalists of **SICE Annual Conference 2022 Young Author's Award**.

[21] A. Li, M. Ogura, Y. Tsunoda, and N. Wakamiya, "Proposal of farthest-agent targeting algorithm with indirect chasing," in *SICE Annual Conference 2022*, 2022, pp. 92–94.

[22] M. Ogura, K. Kobayashi, and K. Sugimoto, "Static output feedback synthesis of time-delay linear systems via deep unfolding," in *17th IFAC Workshop on Time Delay Systems*, 2022, pp. 214–215.

[23] M. Kishida and M. Ogura, "Temporal deep unfolding for nonlinear stochastic optimal control," in *7th International Conference on Advances in Control & Optimization of Dynamical Systems*, 2022, pp. 908–913.

[24] M. Ogura and N. Wakamiya, "Model predictive control of fish schooling model with reduced-order prediction model," in *60th IEEE Conference on Decision and Control*, 2021, pp. 4115–4120.

[25] M. Kishida and M. Ogura, "Temporal deep unfolding for nonlinear maximum hands-off control," in *SICE Annual Conference 2021*, 2021, pp. 1007–1010.

[26] K. Kobayashi, M. Ogura, T. Kobayashi, and K. Sugimoto, "Deep unfolding-based output feedback control design for linear systems with input saturation," in *SICE International Symposium on Control Systems 2021*, 2021, pp. 33–39.

[27] M. Kumazaki, M. Ogura, and T. Tachibana, "Service chain construction with efficient VNF sharing based on model predictive control," in *2020 International Conference on Emerging Technologies for Communications*, 2020, pp. M1–5.

[28] K. Sugimoto, M. Ogura, K. Hanada, and T. Aihara, "Sampled-data suboptimal state estimation over lossy networks," in *52nd ISCIE International Symposium on Stochastic Systems Theory and Its Applications*, 2020, pp. 79–83.

[29] C. Zhao, M. Ogura, and K. Sugimoto, "Finite-time control of discrete-time positive linear systems via convex optimization," in *SICE Annual Conference 2020*, 2020, pp. 1230–1235.

[30] Y. Abe, M. Ogura, H. Tsuji, A. Miura, and S. Adachi, "Resource and network management for satellite communications systems: a chance-constrained approach," in *21st IFAC World Congress*, 2020, pp. 3304–3309.

[31] T. Kimura and M. Ogura, "Distributed collaborative 3D-deployment of UAV base stations for on-demand coverage," in *IEEE International Conference on Computer Communications 2020*, 2020, pp. 1748–1757. Acceptance rate **19.8 percent**.

[32] M. Aida, C. Takano, and M. Ogura, "On the fundamental equation of user dynamics and the structure of online social networks," in *NetSci-X 2020*, 2020, pp. 155–170.

[33] M. Ogura, W. Mei, and K. Sugimoto, "Upper-bounding dynamics on networked synergistic susceptible-infected-susceptible model," in *SICE Annual Conference 2019*, 2019, pp. 1430–1431.

[34] M. Ogura, M. Kishida, K. Hayashi, and J. Lam, "Geometric programming for optimizing stability of distributed power control algorithms," in *SICE Annual Conference 2019*, 2019, pp. 679–680.

[35] M. Ogura, M. Kishida, and A. Yassine, "Optimizing product development projects under asynchronous and aperiodic system-local interactions," in *21st International DSM Conference*, 2019, pp. 97–106.

[36] M. Ogura, M. Kishida, K. Hayashi, and J. Lam, "Resource allocation for robust stabilization of Foschini-Miljanic Algorithm," in *2019 American Control Conference*, 2019, pp. 4030–4035.

[37] M. Kumazaki, M. Ogura, and T. Tachibana, "VNF management with model predictive control for multiple service chains," in *IEEE International Conference on Consumer Electronics – Taiwan*, 2019.

[38] T. Tadenuma, M. Ogura, and K. Sugimoto, "Sampled-data state observation over lossy networks under round-robin scheduling," in *5th IFAC Conference on Analysis and Control of Chaotic Systems*, 2018, pp. 197–202. **Young Author Award Finalist**.

[39] W. Mei and M. Ogura, "Instability analysis of Markov jump linear systems by spectral optimization," in *SICE Annual Conference 2018*, 2018, pp. 419–422.

[40] M. Ogura, J. Wan, and S. Kasahara, "Model predictive control for energy-efficient operation of data centers with cold aisle containments," in *6th IFAC Conference on Nonlinear Model Predictive Control*, 2018, pp. 241–246.

[41] M. Ogura and J. Harada, "Resource allocation for containing epidemics from temporal network data," in *23rd International Symposium on Mathematical Theory of Networks and Systems*, 2018, pp. 537–542.

[42] M. Ogura, J. Tagawa, and N. Masuda, "Distributed agreement on activity driven networks," in *2018 American Control Conference*, 2018, pp. 4147–4152.

[43] X. Chen, M. Ogura, K. R. Ghusinga, A. Singh, and V. M. Preciado, "Semidefinite bounds for moment dynamics: Application to epidemics on networks," in *56th IEEE Conference on Decision and Control*, 2017, pp. 2448–2454.

[44] M. Ogura and V. M. Preciado, "Katz centrality of Markovian temporal networks: analysis and optimization," in *2017 American Control Conference*, 2017, pp. 5001–5006.

[45] M. Wakaiki, M. Ogura, and J. P. Hespanha, "Linear quadratic control for sampled-data systems with stochastic delays," in *2017 American Control Conference*, 2017, pp. 1978–1983.

[46] M. Ogura and V. M. Preciado, "Efficient containment of exact SIR Markovian processes on networks," in *55th IEEE Conference on Decision and Control*, 2016, pp. 967–972. **29**.

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[63] M. Ogura and C. F. Martin, "Stability of switching systems and generalized joint spectral radius," in *2013 European Control Conference*, 2013, pp. 3185–3190.

[64] M. Ogura and C. F. Martin, "Stochastic properties of switched Riccati differential equations," in *51st IEEE Conference on Decision and Control*, 2012, pp. 1319–1324.

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Invited and Hourly Talks

[1] "Swarm control: From simple local rules to collective behavior," *Global Faculty Week, Kaunas University of Technology*, 2026.

[2] "From Local Interventions to Scalable System-Level Effects," *De La Salle University*, 2026.

[3] "Shepherding Dynamics: A Testbed for Nonlinear and Information-Limited Multi-Agent Control," *HKU-UM Workshop on Control Theory and Applications*, 2025.

[4] "Towards Operational and Scalable Control Strategies for Complex Systems," *Jiangnan University*, 2025.

[5] "Toward Control Theory for Complex Systems: From Epidemics to Swarms and Weather," *IDEC Institute, Hiroshima University*, 2025.

[6] "Shepherding as a paradigm for swarm control," *2024 IEEE International Conference on Cyborg and Bionic Systems*, 2024.

[7] "Enhancing Control System Design through Deep Unfolding: A Systematic Approach," *University of Hawaii at Manoa*, 2023.

[8] "Perspectives on Artificial Intelligence," *2nd Japanese-Canadian Frontiers of Science (JCFoS) Symposium*, 2023.

[9] "When deep unfolding meets control engineering," *37th International Technical Conference on Circuits\Systems, Computers and Communications*, 2022.

[10] "Optimization of positive linear systems via geometric programming," *Shenzhen University*, 2019.

[11] "Optimization of positive linear systems via geometric programming," *Guandong University of Technology*, 2019.

[12] "Synthesis of positive linear systems by geometric programming," *University of Hong Kong*, 2019.

[13] "Network epidemiology and control theory," *University of Hong Kong*, 2018.

[14] "Analysis and control of spreading processes over complex networks," *Washington State University*, 2017.

[15] "Analysis and control of spreading processes over complex networks," *Tokyo University of Agriculture and Technology*, 2016.

[16] "Stability analysis of switched linear systems with non-traditional switching signals," in *GRASP special seminar*, University of Pennsylvania, 2014.

[17] "Mean stability of switched linear systems," *Université Catholique de Louvain*, 2013.

TEACHING ACTIVITIES

Hiroshima University

- Reinforcement Learning (2025–present)
- Introduction to Artificial Intelligence (2024–present)
- Calculus II (2024–present)
- Artificial Intelligence and Machine Learning (2024)

Osaka University

- Basic Theory of Bio-networks (2021–present)
- Informatics for Engineering Science 1 (2020–present)
- Fundamentals of Information and Computer Sciences (2020–present)
- Bio-network Engineering (2020–present)

Nara Institute of Science and Technology

- Machine Learning and Intelligent Control (Spring 2019)
- Advanced Intelligent System Control (Spring 2017, 2018)

University of Pennsylvania

Co-lecturer:

- Convex Optimization in Systems and Control (Fall 2015)

Texas Tech University

Graduate Part-Time Instructor:

- Calculus II (Summer 2014, Spring 2014, Spring 2013)
- Calculus I (Summer 2013, Fall 2012)
- Trigonometry (Fall 2011)
- College Algebra (Fall 2013, Spring 2012)

Teaching Assistant:

- Advanced Calculus (Summer 2012)
- Linear Algebra (Summer 2012)
- Higher Mathematics for Engineers and Scientists I (Summer 2011)

Kyoto School of Computer Science

Lecturer:

- Control Engineering (Fall 2009, Fall 2008)
- Electrical Circuits (Spring 2008)
- Data Structures (Spring 2008)
- Numerical Analysis (Spring 2010, Spring 2009)

Kyoto University

Teaching Assistant:

- Modern Control Theory (Fall 2009, Fall 2008)

PROFESSIONAL SERVICE

- Program Committee Member, Sixth International Symposium on Complex Systems
- Associate Editor, 2026 SICE International Symposium on Control Systems
- Organized Invited Sessions Chair, SICE Festival 2026 with Annual Conference
- Program Committee Member, 14th International Conference on Complex Networks and their Applications
- Associate Editor, The Journal of Engineering (2025–present)
- Technical Comittee Member, Technical Comittee on Fully Actuated System Theory and Applications (2025–present)
- International Program Committee Secretary, 8th IFAC Conference on Nonlinear Model Predictive Control
- Publication Co-Chair, 8th IFAC Conference on Nonlinear Model Predictive Control
- Program Committee Member, 13th International Conference on Complex Networks and their Applications

- Associate Editor, 10th Indian Control Conference
- External Reviewer, Research Grants Council of Hong Kong (2023, 2024, 2026)
- Reviewer, French National Research Agency (2023)
- Program Committee member, 12th International Conference on Complex Networks and their Applications
- Registration Chair, 10th IFAC Symposium on Robust Control Design
- International Advisory Committee Member, 2022 International Conference on Information Technology
- Associate Editor, IET Control Theory & Applications (2022–present)
- Associate Editor, Franklin Open (2022–present)
- Program Committee member, 11th International Conference on Complex Networks and their Applications
- Program Committee member, 10t International Conference on Complex Networks and their Applications
- Technical Program Comittee Member, 7th IEEE International Conference on Network Softwarization
- Associate Editor, Journal of The Franklin Institute2020–present
- Program & Steering Committee Member, 52nd ISCIE International Symposium on Stochastic Systems Theory and Its Applications
- Local Arrangements Vice Chair, SICE Annual Conference 2018
- Technical Comittee Member, Technical Committee 1.5. Networked Systems, International Federation of Automatic Control (2015–present).
- Associate Editor, 5th IFAC Workshop on Distributed Estimation and Control in Networked Systems

Journal Reviewer

Advanced Robotics; Annual Reviews in Control; Automatica; Applied Mathematics and Computation; Applied Network Science; Applied Soft Computing; Asian Journal of Control; Chaos, Solitons and Fractals; Computer Communications; Developments in the Built Environment; Discrete Mathematics; Engineering Applications of Artificial Intelligence; Entropy; European Journal of Control European Physical Journal B; Expert Systems With Applications; Foundations of Computational Mathematics; IEEE Access; IEEE/ACM Transactions on Computational Biology and Bioinformatics; IEEE/CAA Journal of Automatica Sinica; IEEE Circuits and Systems Magazine; IEEE Control Systems Letters; IEEE Intelligent Systems; IEEE Internet of Things Journal; IEEE Open Journal of Vehicular Technology; IEEE Signal Processing Letters; IEEE Transactions on Automatic Control; IEEE Transactions on Automation Science and Engineering; IEEE Transactions on Big Data; IEEE Transactions on Circuits and Systems I: Regular Papers; IEEE Transactions on Circuits and Systems II: Express Briefs; IEEE Transactions on Cognitive and Developmental Systems; IEEE Transactions on Computational Social Systems; IEEE Transactions on Control of Network Systems; IEEE Transactions on Cybernetics; IEEE Transactions on Fuzzy Systems; IEEE Transactions on Industrial Electronics; IEEE Transactions on Industrial Informatics; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Neural Networks and Learning Systems; IEEE Transactions on Signal and Information Processing over Networks; IEEE Transactions on Signal Processing; IEEE Transactions on Systems, Man and Cybernetics: Systems; IEEE Transactions on Pattern Analysis and Machine Intelligence; IEEE Transactions on Power Systems; IEEE Transactions on Mobile Computing; IET Control Theory & Applications; International Journal of Systems Science; International Journal of Robust and Nonlinear Control; Internet Technology Letters; Journal of Aerospace Engineering; Journal of

Biological Dynamics; Journal of Building Engineering; Journal of Computational Science; e Journal of Systems Science and Complexity; Journal of the Franklin Institute; Mathematics and Computers in Simulation; Mathematics of Control, Signals, and Systems; National Science Review; Neural Networks; Neurocomputing; Nonlinear Analysis: Hybrid Systems; Nonlinear Dynamics; Operations Research and Decisions; Optimal Control, Applications and Methods; Philosophical Transactions of the Royal Society A; Physica A; Physics Letters A; PLOS Computational Biology; Science China Information Sciences; SIAM Journal on Control and Optimization; SICE Journal of Control, Measurement, and System Integration; Stochastics and Dynamics; Systems and Control Letters; Research in Engineering Design.

Masaki Ogura, October 2026